FX SETTINGS HELP

Color Matrix FX

An advanced 4 x 5 RGBA matrix for channel remapping, inversion, brightness, contrast and Adobe Animate-style color effects.

Find it in OpenToonz

FX Schematic > Add FX > Image Adjust > Color Matrix. Connect a source image or column to Source and edit its settings. All 20 controls can be keyframed.

How the matrix works

Each output row (Red, Green, Blue, Alpha) mixes four input channels and adds an independent Offset. The default is an identity matrix, which leaves the input unchanged.

OUTPUT	RED	GREEN	BLUE	ALPHA	OFFSET
Red	1	0	0	0	0
Green	0	1	0	0	0
Blue	0	0	1	0	0
Alpha	0	0	0	1	0

```
Red' = red_red*R + red_green*G + red_blue*B + red_alpha*A + red_offset/255
```

Ranges and units

Mixing coefficients range from -16 to +16; channel offsets range from -1024 to +1024 in 8-bit units. An offset of +30 adds about 30/255 to a normalized channel value. Output RGB is premultiplied by alpha.

Getting started

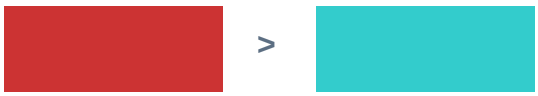
1. Pick an output row. 2. Edit input contributions. 3. Set Offset for a constant addition or subtraction. 4. Animate parameters using regular FX keyframes.

EXAMPLES

Practical color recipes

Begin at the identity defaults. Change only the specified controls unless a recipe replaces a whole row.

INVERT RGB



BRIGHTEN +30

INVERT RGB

Set red_red, green_green, blue_blue = -1 and red_offset, green_offset, blue_offset = +255. Alpha is unchanged.

BRIGHTNESS

Set red_offset, green_offset, blue_offset = +30 to increase color values on nontransparent source pixels.

CONTRAST

Set red_red, green_green, blue_blue = 1.2 and the three RGB offsets = -25.5. Anchors midgray while increasing contrast; extremes may clip.

BOOST RED

Set red_red = 1.4. This gain exceeds Channel Mixer's maximum coefficient of 1.0.

GRAYSCALE

Replace each RGB output row with [0.2126, 0.7152, 0.0722, 0, 0]. Channel Mixer can also express this nonnegative combination.

HALF OPACITY

Set alpha_alpha = 0.5. This adjustment is also available with Channel Mixer.

Offsets do not create new pixels

Pixels whose source alpha is exactly zero are skipped. Even a positive Alpha Offset will not generate content in fully transparent regions.

COMPARISON & RENDERING

Color Matrix vs. Channel Mixer

Both effects mix RGBA channels, but Color Matrix adds signed coefficients and independent offsets.

CAPABILITY	CHANNEL MIXER	COLOR MATRIX
Channel coefficients	16 in 0..1	16 in -16..+16
Independent offsets	None	4 in -1024..+1024
Negative or >1 gains	No	Yes
FX Intensity	Present (global blend)	Not in PR #272
Keyframeable controls	Yes	Yes
8/16-bit and float	Yes	Yes
Identity fast path	No	Yes

Rendering details

Premultiplied alpha.
Both effects mix unpremultiplied colors and then premultiply by output alpha. Different rounding can produce small pixel differences.

Transparent areas.
Color Matrix skips pixels with zero input alpha; Alpha Offset alone cannot fill completely transparent regions.

Floating-point/HDR.
Color Matrix clamps alpha into 0..1 and negative RGB to zero while preserving RGB over 1. Channel Mixer does not explicitly clamp float alpha the same way.

FX Intensity.
Channel Mixer uses GlobalControllableFx to cross dissolve processed and unprocessed images below 100% intensity. Color Matrix currently does not include that slider.

Which one should I use?

Use Channel Mixer for nonnegative channel blends with Intensity. Use Color Matrix for inversion, offsets, contrast shifts, gains above 1 and imported Adobe Animate color filters.

Original FX implementation: @doangelshavewings, Flash Import PR #267.
Standalone Color Matrix extraction: OT-Dev PR #272.